

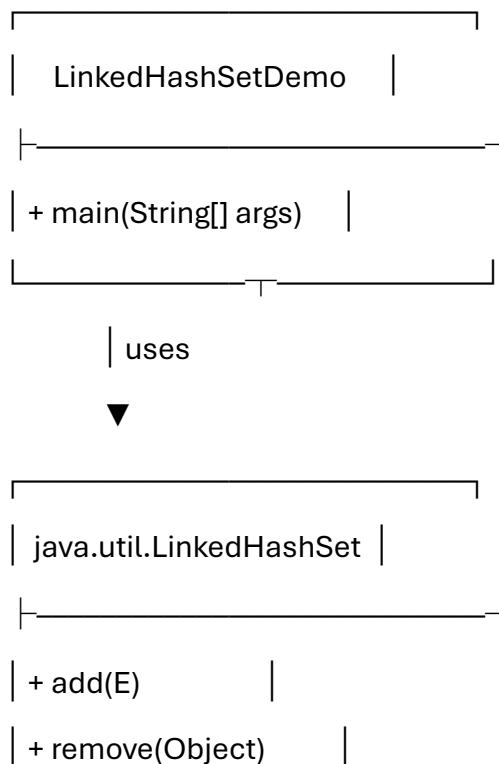
Aim

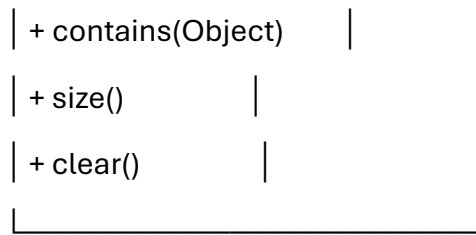
To write a Java program to demonstrate the important methods of the **LinkedHashSet** class, such as add(), remove(), contains(), size(), and clear().

Algorithm

1. Start the program.
2. Import the java.util.LinkedHashSet class.
3. Create a LinkedHashSet object.
4. Use add() to insert elements.
5. Display the elements in insertion order.
6. Use contains() to check whether an element exists.
7. Use remove() to remove an element.
8. Use size() to find the number of elements.
9. Use clear() to remove all elements.
10. Display the results.
11. Stop the program.

UML Diagram





Code:

```
package Observations;
```

```
import java.util.LinkedHashSet;
```

```
public class LinkedHashSetClass {
```

```
    public static void main(String[] args) {
```

```
        // Creating a LinkedHashSet
```

```
        LinkedHashSet<String> set = new LinkedHashSet<>();
```

```
        // add()
```

```
        set.add("Apple");
```

```
        set.add("Banana");
```

```
        set.add("Mango");
```

```
        set.add("Orange");
```

```
        set.add("Apple"); // Duplicate is not added
```

```
        System.out.println("LinkedHashSet after add(): " + set);
```

```
        // contains()
```

```
        System.out.println("Contains Banana: " + set.contains("Banana"));
```

```
        // remove()
```

```

set.remove("Mango");

System.out.println("LinkedHashSet after remove(): " + set);

// size()

System.out.println("Size of LinkedHashSet: " + set.size());

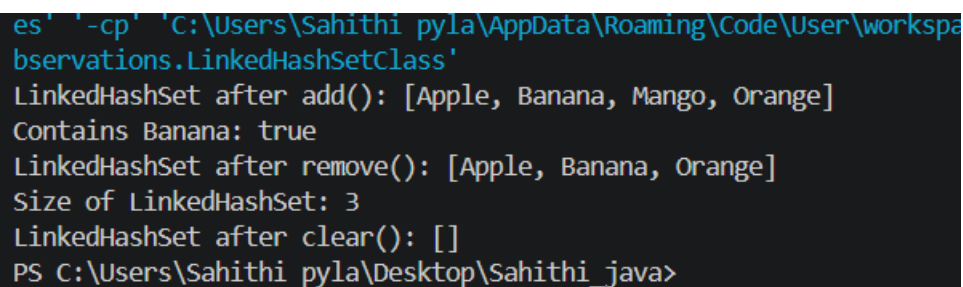
// clear()

set.clear();

System.out.println("LinkedHashSet after clear(): " + set);
}
}

```

Output:



```

es' '-cp' 'C:\Users\Sahithi pyla\AppData\Roaming\Code\User\workspa
bservations.LinkedHashSetClass'
LinkedHashSet after add(): [Apple, Banana, Mango, Orange]
Contains Banana: true
LinkedHashSet after remove(): [Apple, Banana, Orange]
Size of LinkedHashSet: 3
LinkedHashSet after clear(): []
PS C:\Users\Sahithi pyla\Desktop\Sahithi_java>

```

Result:

Thus, the Java program successfully demonstrates the LinkedHashSet methods add(), remove(), contains(), size(), and clear(), while maintaining the insertion order of elements.